

Practical No: 4

Name:- Danish Altaf Solkar

Enrollment No:- 24111590805

Class:- CO3KB

Subject:- CGR

Aim :- To draw a circle using Bresenham's algorithm

Program :-

```
#include<stdio.h>

#include<conio.h>

#include<dos.h>

#include<graphics.h>

void drawCircle(int xc,int yc,int x,int y)

{

    putpixel(xc+x,yc+y,8);

    putpixel(xc-x,yc+y,4);

    putpixel(xc+x,yc-y,15);

    putpixel(xc-x,yc-y,7);

    putpixel(xc+y,yc+x,13);

    putpixel(xc-y,yc+x,3);

    putpixel(xc+y,yc-x,14);

    putpixel(xc-y,yc-x,9);

}

void circleBres(int xc,int yc,int r)

{

    int x=0,y=r;

    int d=3-2*r;

    drawCircle(xc,yc,x,y);
```

```
while(x<=y)

{

x++;

if(d>0)

{

y--;

d=d+4*(x-y)+10;

}

else

{

d=d+4*x+6;

}

drawCircle(xc,yc,x,y);

delay(150);

}

}

void main()

{

int xc=300,yc=200,r=100;

int gd=DETECT,gm;

initgraph(&gd,&gm,"C:\\TurboC3\\BGI");

circleBres(xc,yc,r);

getch();

closegraph();

}
```

OUTPUT :-

